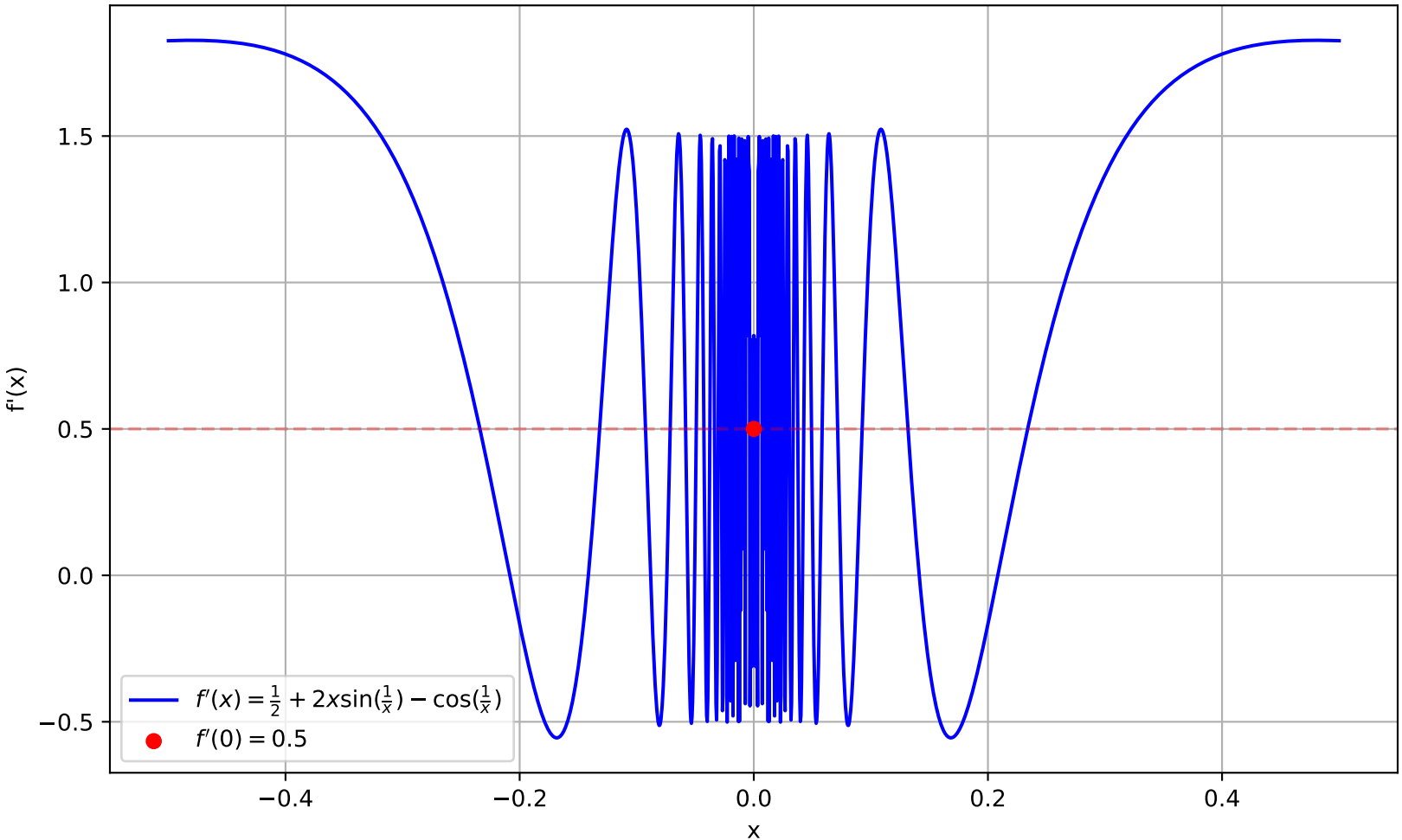
1.5
1.0
0.5
0.0
−0.5
f'(x)
−0.4
−0.2
0.0
0.2
0.4
x
$f'(x) = \frac{1}{2} + 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$
$f'(0) = 0.5$